PAEDIATRIC ORTHO · NEET-SS / FET / INI-SS

bone and joint infection in children

1. Pathophysiology

- **Acute haematogenous osteomyelitis (AHO)** starts in the **metaphysis**: nutrient artery branches end in **sinusoidal loops beside the physis** where blood flow is slow and turbulent, the capillary endothelium has gaps, and there are few phagocytes. Minor trauma may create a haematoma that seeds bacteria.
- **Spread:** pus raises intraosseous pressure, tracks through the **Haversian and Volkmann canals** to the subperiosteal space, and lifts the loose paediatric periosteum (**subperiosteal abscess**). Cortical bone loses its blood supply (from both sides) and dies: **sequestrum**. The elevated periosteum lays down new bone: **involucrum**. Openings in the involucrum through which pus escapes: **cloacae**.
- **Age-related anatomy:**
  - **Under about 18 months: transphyseal vessels** cross the physis, so metaphyseal infection spreads to the **epiphysis and joint** (combined osteomyelitis and septic arthritis; physeal and epiphyseal damage).
  - **Childhood:** the physis is a barrier; infection stays metaphyseal and spreads subperiosteally (thick, loose periosteum).
  - **Adults:** periosteum firmly attached; infection spreads within the medullary canal and into the joint once the physes close.
- **Joints where the metaphysis is intra-articular** (osteomyelitis can rupture into the joint): **proximal femur (hip)**, **proximal humerus (shoulder)**, **proximal radius (elbow)**, **distal lateral tibia (ankle)**.

- **Septic arthritis destroys cartilage quickly:** bacterial toxins and **proteolytic enzymes from neutrophils** (proteoglycan loss within hours, collagen within days). The **hip is the emergency**: raised intracapsular pressure can tamponade the retinacular vessels and cause **AVN of the femoral head**.

2. Organisms by age and setting

Setting	Organisms
Neonate (under 2 months)	**Group B streptococcus**, **Staphylococcus aureus**, Gram-negative enteric bacilli (E. coli), Candida in NICU babies
6 months-4 years	**S. aureus** and **Kingella kingae** (the commonest cause of osteoarticular infection in this age group where PCR is used)
Older children	**S. aureus** (by far commonest at all ages), group A streptococcus
Adolescents (sexually active)	Neisseria gonorrhoeae (septic arthritis, tenosynovitis, skin pustules)
Sickle cell disease	**Salmonella** is characteristic (bowel infarcts allow bacteraemia), S. aureus also common
Puncture through a sneaker	**Pseudomonas aeruginosa** (osteochondritis of the foot)
Unvaccinated child	**Haemophilus influenzae type b** (now rare)
Cat or dog bite	Pasteurella multocida
Chronic granulomatous disease	Aspergillus, Serratia, Burkholderia
India (endemic)	**Mycobacterium tuberculosis** in subacute cases, dactylitis, monoarthritis

- **MRSA** (community-acquired, especially **Panton-Valentine leukocidin (PVL)-positive** strains): severe sepsis, **multifocal disease**, **deep vein thrombosis and septic pulmonary emboli**, necrotising myositis, more surgery.
- **Kingella kingae:** Gram-negative coccobacillus in the oropharynx of toddlers, often after a viral upper respiratory infection. **Mild illness, low-grade or no fever, normal or mildly raised CRP and WBC.** It grows poorly on standard media: **inoculate blood-culture bottles** with joint fluid and use **PCR**. Treat with a cephalosporin (resistant to clindamycin, glycopeptides).

3. Clinical features

- **AHO:** fever, malaise, **localised metaphyseal tenderness**, pseudoparalysis (refusal to use the limb), limp; swelling and erythema later.
- **Septic arthritis:** severely painful joint, **refusal to bear weight**, held in the position of maximum capsular volume (**hip: flexion, abduction and external rotation**), pain on any passive movement (log-roll), fever.
- **Neonates: few systemic signs**; irritability, pseudoparalysis, poor feeding; **multifocal** disease is common; high risk of joint destruction.
- **Psoas abscess, pyomyositis, discitis and sacroiliitis** can mimic septic hip: discitis in toddlers presents with refusal to walk or sit; pyomyositis (S. aureus, tropical) with a tender, woody muscle.

4. Investigations

- **Blood:** WBC, **ESR** (rises over days, falls slowly over weeks), **CRP** (rises within 6-8 hours, peaks at 48 hours, **normalises within about a week of effective treatment**: the best

marker of response), **blood cultures** (positive in about 30-50%).

- **Kocher criteria (septic arthritis vs transient synovitis of the hip): non-weight-bearing**, **temperature over 38.5 C**, **ESR over 40 mm/h**, **WBC over 12,000/mm3**.

Criteria present	Predicted probability of septic arthritis
0	about 0.2%
1	about 3%
2	about 40%
3	about 93%
4	about 99.6%

  - **Caird** added **CRP over 2 mg/dL** (20 mg/L) as an independent predictor; **fever over 38.5 C is the best single predictor** in that study. Later validation studies showed lower accuracy in other populations: use the criteria as a guide, not a rule.

- **Imaging:**
  - **Plain radiographs:** soft-tissue swelling and loss of fat planes early; **bone changes (lysis, periosteal reaction) appear only after about 10-14 days** (30-50% loss of bone mineral needed). Baseline and to exclude differentials.
  - **Ultrasound: hip effusion** (absence of an effusion makes septic hip very unlikely), subperiosteal collections; guides aspiration.
  - **MRI** is the **investigation of choice for osteomyelitis**: marrow oedema (T1 low, T2/STIR high), subperiosteal and intraosseous abscess, physeal involvement, adjacent septic arthritis, pyomyositis. Contrast helps define abscesses.

    - Bone scan (Tc-99m) if the site is unclear or multifocal disease is suspected; can be cold early in severe infection (compromised blood flow).
- **Aspiration** (joint or subperiosteal pus) under ultrasound or fluoroscopy before antibiotics if it does not delay treatment: Gram stain, culture (plus blood-culture bottle and PCR for Kingella), cell count. **Synovial WBC over about 50,000/mm3 with over 75-90% neutrophils** suggests septic arthritis (lower counts in Kingella and early infection).

5. Treatment

Antibiotics

- Start **empirical IV antibiotics immediately after cultures**: **anti-staphylococcal** cover (cefazolin or cloxacillin/flucloxacillin; **add vancomycin or clindamycin where MRSA is prevalent**; add cefotaxime/ceftriaxone in neonates and where Gram-negatives or Kingella are possible).
- **Short IV course with early oral switch** (current evidence, e.g. the Finnish trials of Peltola et al.): switch to high-dose oral antibiotics after about **2-4 days** once the child is clinically improving (afebrile, less pain) and **CRP is falling** (about 50% reduction).
- **Total duration: septic arthritis about 2-3 weeks**; **uncomplicated AHO about 3-4 weeks**; longer (6 weeks or more) for MRSA, complicated disease, neonates, pelvic and spinal infection, slow response.

Surgery

- **Septic arthritis of the hip is a surgical emergency: arthrotomy and thorough washout** (anterior approach; some use arthroscopy) as soon as possible. Other joints (knee,

ankle, elbow) may be managed by arthroscopic washout or repeated aspiration in selected cases.

- **Osteomyelitis: operate** for a **subperiosteal or intraosseous abscess** (drill the cortex, drain the pus), **no clinical improvement after 48-72 hours** of appropriate antibiotics, sequestrum, or concomitant septic arthritis.
- Immobilise briefly for comfort (splint); early mobilisation once pain allows. Watch for **pathological fracture** in extensive osteomyelitis (protect with a cast).

6. Complications

- **Physeal arrest** leading to limb-length discrepancy or angular deformity.
- **AVN of the femoral head** and **hip dislocation or destruction** after septic hip (the neonatal destroyed hip is **Tom Smith arthritis**; Hunka and Choi classifications of sequelae).
- **Chronic osteomyelitis**, **pathological fracture**, recurrence.
- **DVT and septic emboli** (PVL-MRSA), sepsis and multi-organ failure.
- **Joint stiffness, early arthritis**, cartilage loss.

7. Subacute and chronic osteomyelitis

- **Subacute osteomyelitis (Brodie abscess):** low virulence or partial treatment; insidious pain, few systemic signs, often normal markers.
  - Radiograph: well-defined metaphyseal (or epiphyseal) lucency with a **sclerotic rim**, sometimes crossing the physis.
  - **Differentials: osteoid osteoma, Langerhans cell histiocytosis, chondroblastoma (epiphysis), Ewing sarcoma (diaphysis).**

  - **Roberts classification** of subacute osteomyelitis (types by location and appearance); biopsy and curettage when the diagnosis is uncertain.
- **Chronic osteomyelitis: sequestrum, involucrum, cloaca, sinus**.
  - **Treatment: sequestrectomy and debridement** to healthy bleeding bone (**paprika sign**), but **wait until the involucrum is strong enough** to support the limb (otherwise fracture or segmental loss); dead-space management (antibiotic beads or cement, bone graft later), soft-tissue cover, prolonged antibiotics guided by deep cultures.
  - Long-standing sinus: watch for **Marjolin ulcer (squamous cell carcinoma)** in adults.
  - **Cierny-Mader** staging (anatomical type I-IV and host A, B, C) is used in adults.
- **Chronic recurrent multifocal osteomyelitis (CRMO; chronic non-bacterial osteomyelitis):** autoinflammatory, **sterile cultures**, recurrent multifocal bone pain (**medial clavicle**, metaphyses, spine), associated **palmoplantar pustulosis, psoriasis or inflammatory bowel disease** (SAPHO in adults). Whole-body MRI; **NSAIDs** first line, then bisphosphonates or anti-TNF agents; not long-term antibiotics.

8. Special forms

- **Neonatal osteoarticular infection:** NICU babies with umbilical or central lines; multifocal; minimal signs; **combined osteomyelitis and septic arthritis** (transphyseal vessels); high rates of growth disturbance and hip destruction.
- **Discitis (toddlers, about 1-5 years):** the **disc is vascular in young children**. Refusal to walk or sit, back pain, abdominal pain. Mild fever, raised ESR/CRP. **MRI** shows disc space

narrowing with endplate oedema. Treat with **antibiotics (anti-staphylococcal) and rest/brace**; biopsy rarely needed.

- **Pelvic osteomyelitis and sacroiliitis:** buttock, groin or abdominal pain, mimics septic hip or appendicitis; FABER test; MRI.
- **Puncture wound osteochondritis of the foot (Pseudomonas):** usually osteochondral (metatarsal heads, calcaneus); needs surgical debridement plus anti-pseudomonal antibiotics (ciprofloxacin, ceftazidime).
- **Sickle cell crisis vs osteomyelitis:** both cause bone pain, fever and raised markers. Infarction is far commoner; osteomyelitis is suggested by persistent fever, a single site, positive blood cultures, and fluid collections on ultrasound/MRI. **Salmonella cover** (e.g. ceftriaxone or ciprofloxacin) in empirical treatment. **Dactylitis (hand-foot syndrome)** in infants is infarction.
- **Gonococcal arthritis** (adolescents): migratory polyarthralgia, tenosynovitis, pustular skin lesions; ceftriaxone.
- **Lyme arthritis** (endemic regions): large-joint effusion with moderate pain and less fever; serology.
- **Osteoarticular tuberculosis in children (India):**
  - Insidious monoarthritis (hip, knee), **"spina ventosa"** (tuberculous dactylitis: expanded phalanx or metacarpal), spinal TB (see Spine notes).
  - **Phemister triad:** juxta-articular osteopenia, peripheral erosions, gradual joint space narrowing.
  - Diagnosis: histology (caseating granulomas), **GeneXpert/CBNAAT** (also detects rifampicin resistance), culture.
  - **Multidrug therapy** for about 12 months under the national programme; surgery for abscesses, debridement, deformity.

9. Differential diagnosis of the irritable hip in a child

Condition	Key features
Transient synovitis	Commonest cause of acute hip pain (3-8 years), often after a viral illness, mild or no fever, able to bear some weight, near-normal markers; resolves in 1-2 weeks with rest and NSAIDs
Septic arthritis	Toxic, febrile, refuses weight bearing, raised markers (Kocher, CRP)
Perthes disease	4-8 years, insidious limp, mild pain, normal markers, radiograph or MRI changes
SCFE	Adolescent, often obese, Drehmann sign
Osteomyelitis (proximal femur, pelvis)	Focal tenderness, MRI
Juvenile idiopathic arthritis	Morning stiffness, more chronic, other joints, uveitis risk
Malignancy (leukaemia, Ewing)	Night pain, systemic features, abnormal blood count, bone lesion

Exam pearl: **Metaphysis = commonest site; S. aureus = commonest organism at all ages; Kingella = 6 months-4 years (PCR, blood-culture bottles); Salmonella = sickle cell; Pseudomonas = sneaker puncture.**

Exam pearl: **MRI is the investigation of choice for osteomyelitis; radiographs lag 10-14 days. CRP is the best marker to monitor response. Septic hip = emergency arthrotomy.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.